HW #4

1)

$\cancel{\vec{A} - 2.00\vec{B} + \frac{1}{3}\vec{C} = 0}$

$\cancel{\hat{i}: 2.00 + 2.00 \cdot 4.00 + \frac{1}{3}\vec{C}\hat{i} = 0}$

$\cancel{\hat{j}: -3.00}$

$\cancel{\hat{k}:}$

$\vec{A} - 2.00\vec{B} + \frac{1}{3}\vec{C} = 0$

$\hat{i}: 2.00 + 2.00 \cdot 4.00 + \frac{1}{3}C\hat{i} = 0$

$10.00 + \frac{1}{3}C\hat{i} = 0$

$C\hat{i} = -30.00$

$\hat{j}: -3.00 - 2.00 \cdot 1.00 + \frac{1}{3}C\hat{j} = 0$

$-5.00 + \frac{1}{3}C\hat{j} = 0$

$C\hat{j} = 15.00$

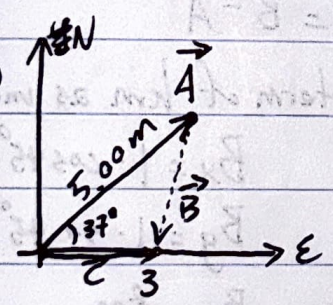
$\hat{k}: 1.00 - 2.00 \cdot -5.00 + \frac{1}{3}C\hat{k} = 0$

$-9.00 + \frac{1}{3}C\hat{k} = 0$

$C\hat{k} = 27.00$

$\vec{C} = -30.00\hat{i} + 15.00\hat{j} + 27.00\hat{k}$

2)



$A_x = 5.00m \cdot \cos(37°)$

$= 3.99\hat{i}$

$A_y = 5.00m \cdot \sin 37°$

$= 3.01\hat{j}$

$\vec{A} = 3.99\hat{i} + 3.01\hat{j}$

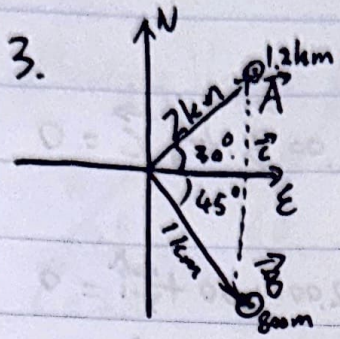
$\vec{C} = 3.00\hat{i} + 0.00\hat{j}$

$\vec{B} = \vec{C} - \vec{A}$

$= (3.00 - 3.99)\hat{i} + (0.00 - 3.01)\hat{j}$

$= -.99\hat{i} + -3.01\hat{j}$

3.



$\vec{C} = \vec{A} - \vec{B}$

$A_x = 2km \cdot \cos 30° = 1.73\ km$

$A_y = 2km \cdot \sin 30° = 1\ km$

$A_z = 1.2km$

$B_x = 1km \cdot \cos 45° = .71\ km$

$B_y = 1\ km \cdot \sin 45° = .71\ km$

$B_z = .8\ km$

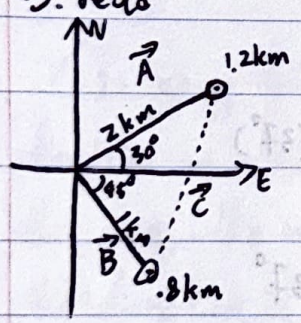
$\vec{C} = \vec{A} - \vec{B}$

$= (1.73 - .71)\ km_x + (1 - .71)\ km_y + (1.2 - .8)\ km_z$

$= 1.02\ km_x + .29\ km_y + .4\ km_z$

Ignore This Please

3. redo



$\vec{C} = \text{Final} - \text{initial} = \vec{B} - \vec{A}$

Unit Vectors are in term of km as units

$A_x = 2 \cdot \cos 30° = 1.73\hat{i}$ $\quad$ $B_x = 1 \cdot \cos 45° = .71\hat{i}$

$A_y = 2 \cdot \sin 30° = 1\hat{j}$ $\quad$ $B_y = 1 \cdot \sin 45° = .71\hat{j}$

$A_z = 1.2\hat{k}$ $\quad$ $B_z = 800m = .8km = .8\hat{k}$

$C_x = .71 - 1.73\hat{i}$

$= -1.02\hat{i}$

$C_y = .71 - 1\hat{j}$

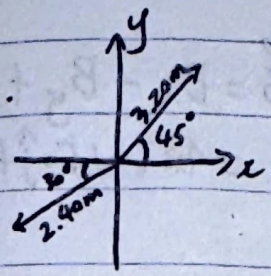
$= -.29\hat{j}$

$C_z = .8 - 1.2\hat{k}$

$= -.4\hat{k}$

$\vec{C} = -1.02\hat{i} + -.29\hat{j} + -.4\hat{k}$

4.



$\vec{A} = A_x + A_y$

$= 3.2 \cdot \cos 45° \hat{i} + 3.2 \cdot \sin 45° \hat{j}$

$= 2.26\hat{i} + 2.26\hat{j}$

$\vec{B} = B_x + B_y$

$= 2.4 \cdot -\cos 30° \hat{i} + 2.4 \cdot -\sin 30° \hat{j}$

$= -2.08\hat{i} + -1.2\hat{j}$

Unit Vectors in this answer ~~uses~~ use "m" as unit

a) $\vec{A} \cdot \vec{B}$

$= A_x B_x + A_y B_y$

$= (2.26 \cdot -2.08)\hat{i} + (2.26 \cdot -1.2)\hat{j}$

$= -4.70\hat{i} + -2.712\hat{j}$

b) $\vec{A} \times \vec{B}$

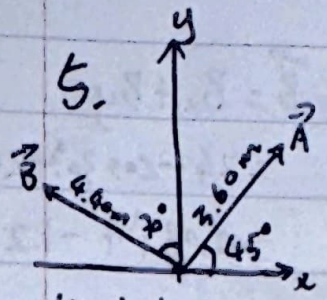
$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2.26 & 2.26 & 0 \\ -2.08 & -1.2 & 0 \end{vmatrix}$$

$$= \hat{i}\begin{vmatrix} 2.26 & 0 \\ -1.2 & 0 \end{vmatrix} - \hat{j}\begin{vmatrix} 2.26 & 0 \\ -2.08 & 0 \end{vmatrix} + \hat{k}\begin{vmatrix} 2.26 & 2.26 \\ -2.08 & -1.2 \end{vmatrix}$$

$= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(-2.712 - -4.7008)$

$= 0\hat{i} + 0\hat{j} + 1.9888\hat{k}$

5.



$\vec{A} = A_x + A_y + A_z$
$= 2.55\hat{i} + 2.55\hat{j} + 0\hat{k}$

$\vec{B} = B_x + B_y + B_z$
$= -4.13\hat{i} + 1.50\hat{j} + 0\hat{k}$

Unit Vectors in this answer are all in term of meters (m)

a) $\vec{A} \cdot \vec{B}$

$= A_xB_x + A_yB_y + A_zB_z$

$= 2.55 \cdot -4.13\hat{i} + 2.55 \cdot 1.50\hat{j} + 0 \cdot 0\hat{k}$

$= -10.5315\hat{i} + 3.825\hat{j} + 0\hat{k}$

b) $\vec{A} \times \vec{B}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2.55 & 2.55 & 0 \\ -4.13 & 1.50 & 0 \end{vmatrix}$$

$$= \hat{i}\begin{vmatrix} 2.55 & 0 \\ 1.50 & 0 \end{vmatrix} - \hat{j}\begin{vmatrix} 2.55 & 0 \\ -4.13 & 0 \end{vmatrix} + \hat{k}\begin{vmatrix} 2.55 & 2.55 \\ -4.13 & 1.50 \end{vmatrix}$$

$= 0\hat{i} + 0\hat{j} + \hat{k}(3.825 - -10.5315)$

$= 0\hat{i} + 0\hat{j} + 14.3565\hat{k}$

6.

Given: ~~A = 5.00 m~~

$\vec{A} = 3.00\hat{i} + 6.00\hat{j} - 2.00\hat{k}$

$B = 5.00\,m$

$\vec{B}$ is perpendicular to $\vec{A}$

$B_z = 0$

$\because$ $\vec{B}$ is perpendicular to $\vec{A}$

$\therefore \vec{A} \cdot \vec{B} = 0$

$\therefore A_x B_x + A_y B_y + A_z B_z = 0$

$\because B_z = 0$

$\therefore A_x B_x + A_y B_y = 0$

$3.00 B_x + 6.00 B_y = 0$

$B_x + 2B_y = 0$

$\because B = 5.00\,m$

$\therefore 5 \cdot \cos(\theta) + 5 \cdot \sin(\theta) = B$

$\therefore 5\cos\theta + 2 \cdot 5\sin\theta = 0$

$\cos\theta + 2\sin\theta = 0$

$2\frac{\sin\theta}{\cos\theta} + 1 = 0$

$2\tan\theta = -1$

$\tan\theta = -\frac{1}{2}$

$\theta = -\tan^{-1}\frac{1}{2}$

$= 26.57^\circ + k180^\circ$

$B_x = 5.00\,m \cdot \cos\theta$

$= 5.00\,m \cdot .894$

$= 4.47\hat{i}$

$B_y = 5.00\,m \cdot \sin\theta$

$= 5.00\,m \cdot 0.447$

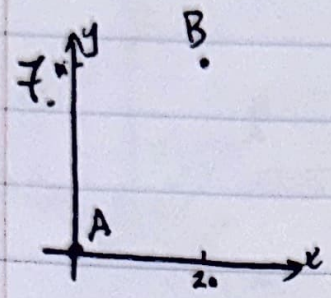
$= 2.24\hat{j}$

$\vec{B} = 4.47\hat{i} + 2.24\hat{j} + 0\hat{k}$

$\vec{C} = -\vec{B}$

$= -4.47\hat{i} + -2.24\hat{j} + 0\hat{k}$

7.



Given : $A(0,0)$, $B(20,35)$

$\Delta t = 5.00s - 0s = 5.00s$

$AV = 15.0\ m/s$, $BV = 30.0\ m/s$

$AV_{direction} = +37°$, $BV_{direction} = +53°$

$\Delta d = B - A$

$= (20 - 0\hat{i}) + (35 - 0\hat{j})$

$= 20\hat{i} + 35\hat{j}$

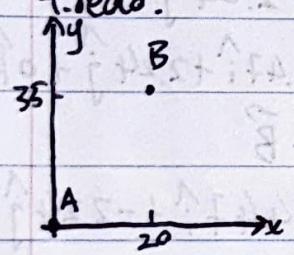
$\Delta V = \cancel{B}BV - AV$

$BV = x\cos 53° + y\sin 53°$

$30\ m/s =$

Please ignore this ...

7.redo.



Given : $A = (0\hat{i}, 0\hat{j})$, $B = (20\hat{i}, 35\hat{j})$

$At = 0s$, $Bt = 5s$; $Av = 15.0\ m/s$, $Bv = 30.0\ m/s$

$Av_{dir} = +37°$, $Bv_{dir} = +53°$

b) $Bv = Bx + By$

$= 30\cos 53° + 30\sin 53°$

$= 18.05\ m/s\ \hat{i} + 23.96\ m/s\ \hat{j}$

$Av = Avx + \cancel{Bvy}\ Avy$

$= 15\cos 37° + 15\sin 37°$

$= 11.98\ m/s\ \hat{i} + 9.03\ m/s\ \hat{j}$

$\Delta V = Bv - Av$

$= (18.05\ m/s\ \hat{i} + 23.96\ m/s\ \hat{j}) - (11.98\ m/s\ \hat{i} + 9.03\ m/s\ \hat{j})$

$= (18.05 - 11.98)\ m/s\ \hat{i} + (23.96 - 9.03)\ m/s\ \hat{j}$

$= 6.07\ m/s\ \hat{i} + 14.93\ m/s\ \hat{j}$

$a_{avg} = \dfrac{6.07\ m/s\ \hat{i} + 14.93\ m/s\ \hat{j}}{5s}$

$= 1.21\ m/s^2\ \hat{i} + 2.99\ m/s^2$

2) $A = (0\hat{i}, 0\hat{j})$

$B = (20\hat{i}, 35\hat{j})$

$\Delta r = (20m - 0m)\hat{i} + (35m - 0m)\hat{j}$

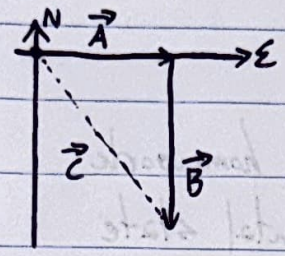
$= 20m\hat{i} + 35m\hat{i}$

$V_{avg} = \frac{20m\hat{i} + 35m\hat{j}}{5s}$

$= \cancel{14m/s\,\hat{i}}$

$= 4\,m/s\,\hat{i} + 7\,m/s\,\hat{j}$

8.



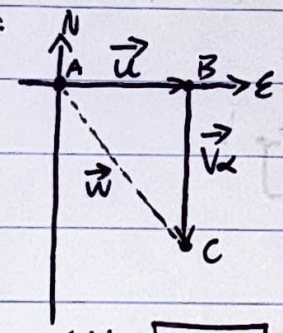
Given: $\vec{A} = 470km\,\hat{i} + 0km\,\hat{j}$

$\vec{B} = 0km\,\hat{i} + 966km\,\hat{j}$

$A_t = .75h \quad B_t = 1.50h$

Please Ignore again... I'm not in a good mental state

redo:



Given: $\vec{u} = 470km\,\hat{i} + 0km\,\hat{j}$

$\vec{V_\alpha} = 0km\,\hat{i} - 966km\,\hat{j}$

$u_t = .75h \quad V_{\alpha t} = 1.50h$

a) $W = \sqrt{u^2 + v_\alpha^2}$

$= \sqrt{470^2 + 966^2}\,km$

$= 1074.27km$

b) $\theta = \tan^{-1}\left(\frac{u}{v_\alpha}\right)$

$= \tan^{-1}\frac{470km}{-966km}$

$= -25.94°$

~~c) $V_{avg} = \frac{\Delta r}{\Delta t} = \frac{r_{total}}{t_{total}}$~~

~~$= \frac{470km + 966km}{.75h + 1.50h}$~~

~~$= 658.22\,km/h$~~

d) $V_u = \frac{470km}{.75h}\hat{i} + 0\,km/h\,\hat{j}$

$= 626.67\,km/h\,\hat{i} + 0\,km/h\,\hat{j}$

$V_{V\alpha} = 0\,km/h\,\hat{i} + \frac{966km}{1.5h}\hat{j}$

$= 0\,km/h\,\hat{i} - 644\,km/h\,\hat{j}$

$V_\theta = \tan^{-1}\frac{V_{avgx}}{V_{avgy}}$

$= -\tan^{-1}\frac{208.89\,km/h\,\hat{i}}{429.33\,km/h\,\hat{j}}$

$= \cancel{44.22°}\ -25.94°$

c) $V_{avg} = \frac{\Delta r}{\Delta t}$

$= \frac{470km}{2.25h}\hat{i} + -\frac{966km}{2.25h}\hat{j}$

$= 208.89\,km/h\,\hat{i} + -429.33\,km/h\,\hat{j}$

e) $S_{avg} = \frac{s_{total}}{t_{total}}$

$= \frac{470\,km + \cancel{649}\,966\,km}{.75h + 1.50h}$

$= 638.22\ km/h$

So sorry for the messy homework time, I am not in a stable mental state recently. I'll try to adjust myself & not do the same next time...

Thanks for your understanding!